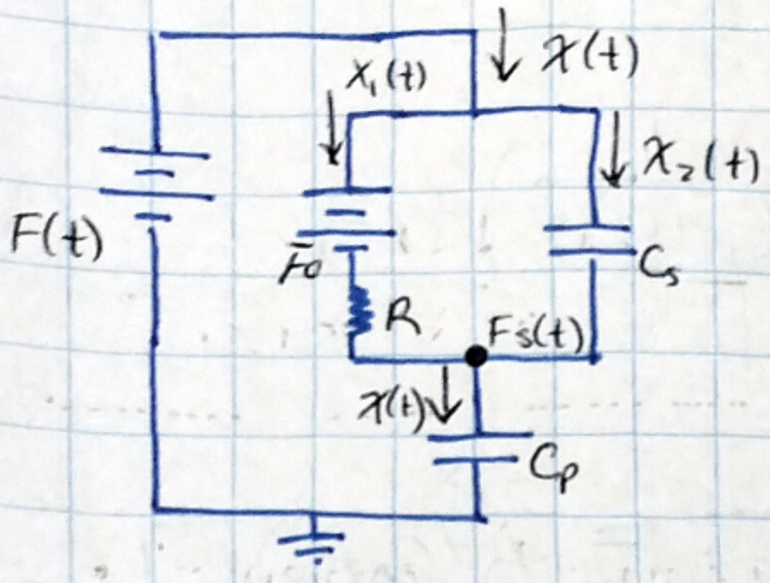
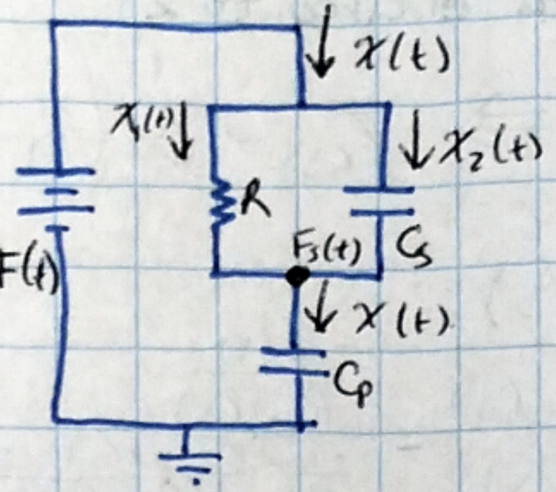
Práctica 3: Modelo y Control del Sistema Musculoesquelético.

→ Transformación de Sistema Mecánico a Circuito Eléctrico:

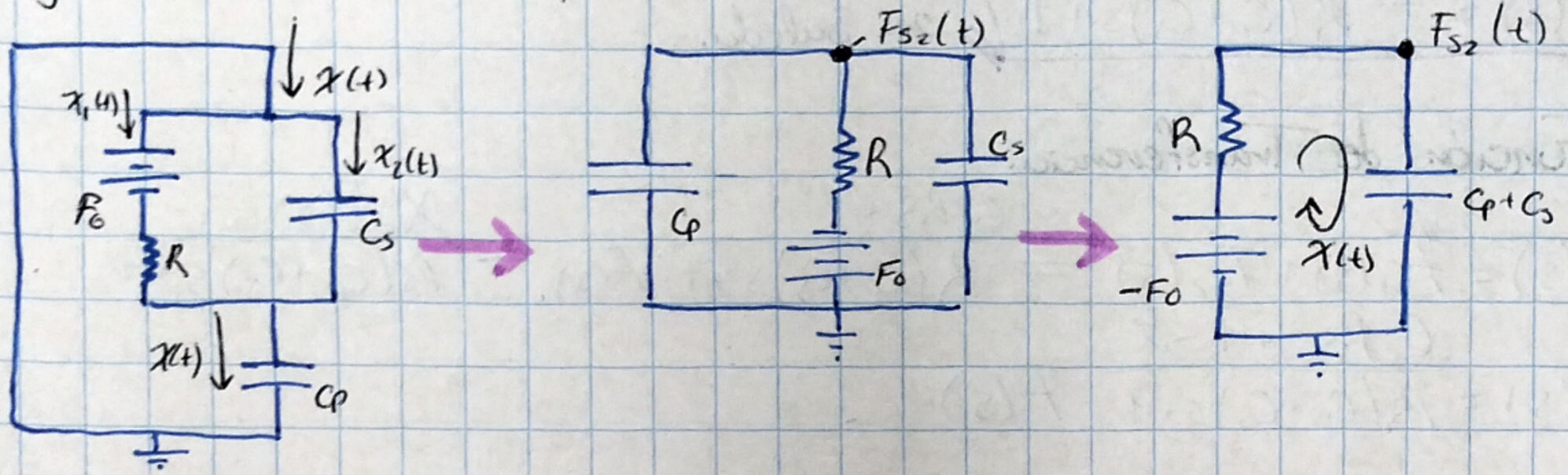


→ Apagando F_0:



Nota: Debido a que el sistema contiene dos fuentes de voltaje, se debe de aplicar el principio de Super posición.

→ Apagando Fuente Principal ($F(t)$):



→ Análisis Matemático:

- Ecuaciones principales para el circuito 1 (Apagando $F_0(t)$).

$$x(t) = x_1(t) + x_2(t) \qquad F_s(t) = F_1(s) + F_2(s).$$

$$\rightarrow x(t) = C_p \cdot \frac{dF_{s_1}(t)}{dt} \qquad \rightarrow x_1(t) = C_s \cdot \frac{d[F(t) - F_{s_1}(t)]}{dt} \qquad \rightarrow x_2 = \frac{F(t) - F_{s_1}(t)}{R}$$

$$\rightarrow C_p \cdot \frac{dF_{s_1}(t)}{dt} = C_s \cdot \frac{d[F(t) - F_{s_1}(t)]}{dt} + \frac{F(t) - F_{s_1}(t)}{R}$$

- Transformada de Laplace para circuito 1.

$$\rightarrow C_p s F_{s_1}(s) = C_s \cdot s[F(s) - F_{s_1}(s)] + \frac{F(s) - F_{s_1}(s)}{R}$$

$$C_p s F_{s_1}(s) = C_s s F(s) - C_s s F_{s_1}(s) + \frac{F(s)}{R} - \frac{F_{s_1}(s)}{R}$$

$$C_p s F_{s_1}(s) + C_s s F_{s_1}(s) + \frac{F_{s_1}(s)}{R} = C_s s F(s) + \frac{F(s)}{R}$$

$$\left(C_p s + C_s s + \frac{1}{R}\right) F_{s_1}(s) = \left(C_s s + \frac{1}{R}\right) F(s) \Rightarrow \frac{R(C_p + C_s)s + 1}{R} F_{s_1}(s) = \frac{C_s R s + 1}{R} F(s)$$

$$F_{s_1}(s) = \frac{\frac{C_s R s + 1}{R}}{\frac{R(C_p + C_s) + 1}{R}}$$

$$\boxed{F_{s_1}(s) = \frac{C_s R s + 1}{R(C_p + C_s)s + 1} F(s)} \quad \text{1era salida.}$$

- Ecuaciones Principales para el circuito 2 (Apagando $F(t)$).

Recordando que: $F_0(t) = \alpha F(t)$.

$$\rightarrow -\alpha F(t) = R \cdot x(t) + \frac{1}{C_p + C_s}\int x(t)\,dt \qquad \rightarrow F_{s2}(t) = \frac{1}{C_p + C_s}\int x(t)\,dt$$

- Transformada de Laplace para el circuito 2.

$$\rightarrow -\alpha F(s) = R X(s) + \frac{X(s)}{(C_p + C_s)s} \Rightarrow F(s) = \frac{-1}{\alpha}\left(R + \frac{1}{(C_p + C_s)s}\right)x(s)$$

$$F(s) = \frac{-1}{\alpha} \cdot \frac{R(C_p + C_s)s + 1}{(C_p + C_s)s}\, X(s)$$

$$\rightarrow F_{s2}(s) = \frac{X(s)}{(C_p + C_s)s} \Rightarrow \frac{F_{s2}(s)}{F(s)} = \frac{\frac{X(s)}{(C_p + C_s)s}}{\frac{-1}{\alpha} \cdot \frac{R(C_p + C_s)s + 1}{(C_p + C_s)s}} = \frac{-\alpha}{R(C_p + C_s)s + 1}$$

$$\underline{F_{s2}(s) = -\frac{\alpha F(s)}{R(C_p + C_s)s + 1}} \quad \text{2da Salida.}$$

- Función de Transferencia.

$$F_s(s) = F_{s1}(s) + F_{s2}(s) = \frac{C_s R s + 1}{R(C_p + C_s)s + 1} \cdot F(s) - \frac{\alpha F(s)}{R(C_p + C_s)s + 1}$$

$$F_s(s) = \frac{C_s R s + 1 - \alpha}{R(C_p + C_s)s + 1}\, F(s)$$

$$\underline{\frac{F_s(s)}{F(s)} = \frac{C_s R s + 1 - \alpha}{R(C_p + C_s)s + 1}} \quad \text{Función de Transferencia.}$$

→ Error en Estado Estacionario.

$$e(s) = \lim_{s \to 0} s F(s)\left(1 - \frac{F_s(s)}{F(s)}\right) = \lim_{s \to 0} s \frac{1}{s}\left(1 - \frac{RC_s s^{\circ} + 1 - \alpha}{R(C_p + C_s)s^{\circ} + 1}\right) = \underline{\alpha}$$

- Estabilidad en Lazo Abierto.

$$(RC_p + RC_s)s + 1 = \emptyset \quad \therefore \quad \underline{\lambda = \frac{1}{-RC_p + RC_s}} \quad \text{El sistema es estable.}$$